

CO4-AT2-Case Study

Course: DSA0103-Object Oriented Programming with C++

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Code:

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4 class Doctor {
5 public:
6     string name;
7     Doctor(string name) {
8         this->name = name;
9     }
10    void display() {
11        cout << "Doctor: " << name << endl;
12    }
13 };
14 class Patient {
15 protected:
16     string name;
17     int age;
18     Doctor doctor;
19 public:
20     Patient(string name, int age, string doctorName)
21         : doctor(doctorName) {
22         this->name = name;
23         this->age = age;
24     }
25     virtual int calculateBill() = 0;
26     virtual void display() {
27         cout << "Patient Name: " << name << endl;
28         cout << "Age: " << age << endl;
29         doctor.display();
30     }
31     virtual ~Patient() {}
32 };
33 class InPatient : virtual public Patient {
34 protected:
35     int roomCharge;
36     int treatmentCost;
```

```
37 public:
38     InPatient(string name, int age, string doctorName,
39               int room, int treatment)
40         : Patient(name, age, doctorName) {
41         roomCharge = room;
42         treatmentCost = treatment;
43     }
44     int calculateBill() override {
45         return roomCharge + treatmentCost;
46     }
47 };
48 class OutPatient : virtual public Patient {
49 protected:
50     int consultationFee;
51 public:
52     OutPatient(string name, int age, string doctorName,
53               int consultation)
54         : Patient(name, age, doctorName) {
55         consultationFee = consultation;
56     }
57     int calculateBill() override {
58         return consultationFee;
59     }
60 };
61 class EmergencyPatient : virtual public Patient {
62 protected:
63     int emergencyCharge;
64 public:
65     EmergencyPatient(string name, int age, string doctorName,
66                      int emergency)
67         : Patient(name, age, doctorName) {
68         emergencyCharge = emergency;
69     }
70     int calculateBill() override {
71         return emergencyCharge;
72     }
73 };
```

```

74 class CriticalPatient : public InPatient, public EmergencyPatient {
75 public:
76     CriticalPatient(string name, int age, string doctorName,
77                     int room, int treatment, int emergency)
78         : Patient(name, age, doctorName),
79           InPatient(name, age, doctorName, room, treatment),
80           EmergencyPatient(name, age, doctorName, emergency) {
81     }
82     int calculateBill() override {
83         return roomCharge + treatmentCost + emergencyCharge;
84     }
85 };
86 int main() {
87     Patient *p;
88     int choice;
89     string name, doctor;
90     int age;
91     cout << "1. InPatient" << endl;
92     cout << "2. OutPatient" << endl;
93     cout << "3. EmergencyPatient" << endl;
94     cout << "4. CriticalPatient" << endl;
95     cout << "Enter choice: ";
96     cin >> choice;
97     cout << "Enter patient name: ";
98     cin >> name;
99     cout << "Enter age: ";
100    cin >> age;
101    cout << "Enter doctor name: ";
102    cin >> doctor;
103    if (choice == 1) {
104        int room, treatment;
105        cout << "Enter room charge: ";
106        cin >> room;
107        cout << "Enter treatment cost: ";
108        cin >> treatment;
109        p = new InPatient(name, age, doctor, room, treatment);
110    }
111    else if (choice == 2) {

```

```

112     int consultation;
113     cout << "Enter consultation fee: ";
114     cin >> consultation;
115     p = new OutPatient(name, age, doctor, consultation);
116 }
117 else if (choice == 3) {
118     int emergency;
119     cout << "Enter emergency charge: ";
120     cin >> emergency;
121     p = new EmergencyPatient(name, age, doctor, emergency);
122 }
123 else if (choice == 4) {
124     int room, treatment, emergency;
125     cout << "Enter room charge: ";
126     cin >> room;
127     cout << "Enter treatment cost: ";
128     cin >> treatment;
129     cout << "Enter emergency charge: ";
130     cin >> emergency;
131     p = new CriticalPatient(
132         name, age, doctor,
133         room, treatment, emergency
134     );
135 }
136 else {
137     cout << "Invalid choice";
138     return 0;
139 }
140 cout << "\n--- Patient Details ---" << endl;
141 p->display();
142 cout << "Total Bill: Rs." << p->calculateBill() << endl;
143 delete p;
144 return 0;
145 }

```

INPUT

```

4
Shreya
15
Reddy
5000
5000
1500

```

Output:

```
1. InPatient
2. OutPatient
3. EmergencyPatient
4. CriticalPatient
Enter choice: Enter patient name: Enter age: Enter doctor name:
Enter room charge: Enter treatment cost: Enter emergency charge:
--- Patient Details ---
Patient Name: Shreya
Age: 15
Doctor: Reddy
Total Bill: Rs.11500
```